10 (a) A stiff metal wire is used to form a rectangular frame PQRS measuring 8.7 cm by 7.2 cm. The frame is open at the top, and is suspended from a sensitive newton meter as shown in Fig. 10.1.

XY is a vertical axis passing through the newton meter and frame PQRS.

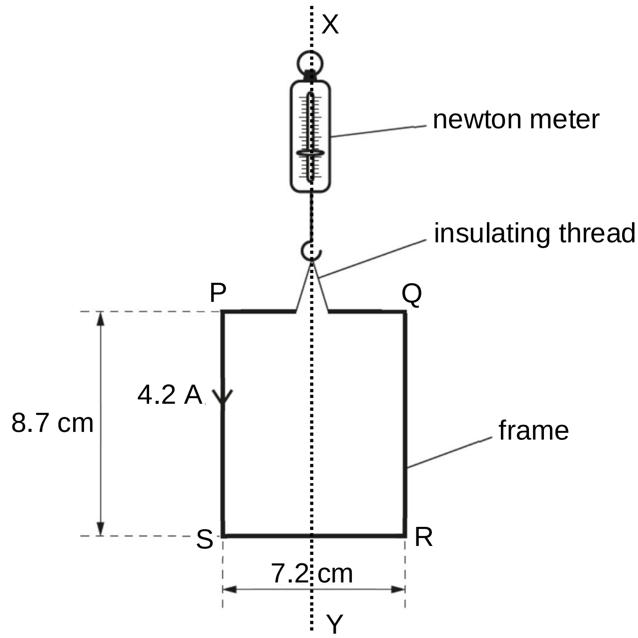

Fig. 10.1

The open ends of the frame are connected to a power supply so that there is a current of 4.2 A in the frame in the direction indicated in Fig. 10.1.

The frame is slowly lowered into a uniform magnetic field of flux density *B* so that side SR is in the field. The magnetic field lines are horizontal and at an angle of 53° to the frame as shown in Fig. 10.2.

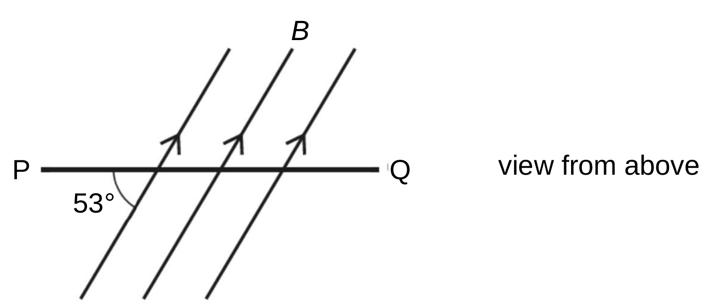

Fig. 10.2

When side SR of the frame first enters the magnetic field, the reading on the newton meter changes by 6.0 mN.

(i) Calculate B.

B = ... T [2]

(ii) State and explain whether the change in the reading on the newton meter is an increase or a decrease.

....

....

.... [2]

(iii) The frame is lowered further so that the vertical sides start to enter the magnetic field.

1. On Fig. 10.2, draw arrows to show the direction of the forces acting on sides PS and QR. [2]

2. The frame is rotated through 90° about the XY axis.

Calculate the magnitude of the torque exerted on the frame in the new orientation when the whole frame is in the magnetic field.

torque = ... N m [3]

3. Suggest what effect the forces will have on the frame eventually.

....

.... [1]

(b) A small solenoid of cross-sectional area 1.4×10^{-3} m^2 is placed inside a larger solenoid of cross-sectional area 4.2×10^{-3} m^2 as shown in Fig. 10.3.

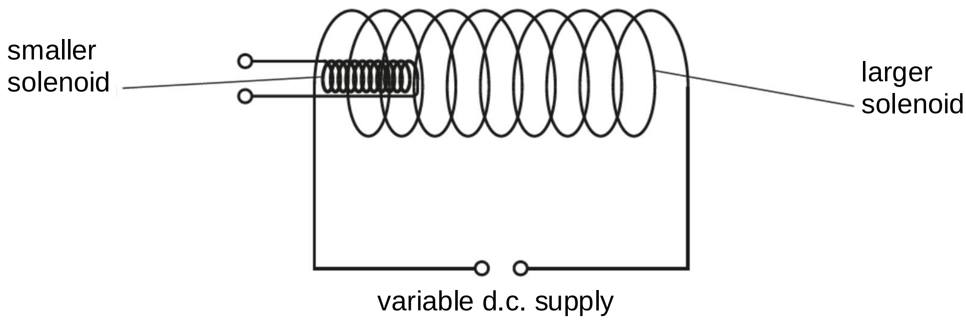

Fig. 10.3

The larger solenoid has 40 turns and is attached to a variable d.c. power supply. The smaller solenoid has 320 turns.

(i) Compare quantitatively the magnetic flux linkage in the two solenoids.

....

.... [2]

(ii) The variation with time t of the magnetic flux density B of the larger solenoid is shown in Fig. 10.4.

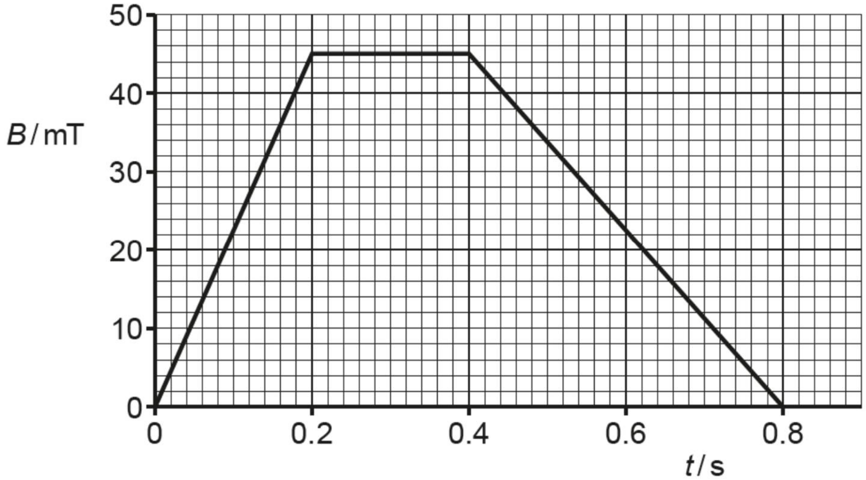

Fig. 10.4

1. Calculate the magnitude of the e.m.f. induced in the smaller solenoid from $t = 0$ to $t = 0.20$ s.

e.m.f. = ... V [2]

2. On Fig. 10.5, sketch a graph to show the variation with time t of the e.m.f. induced E in the smaller solenoid for $t = 0$ to $t = 0.80$ s.

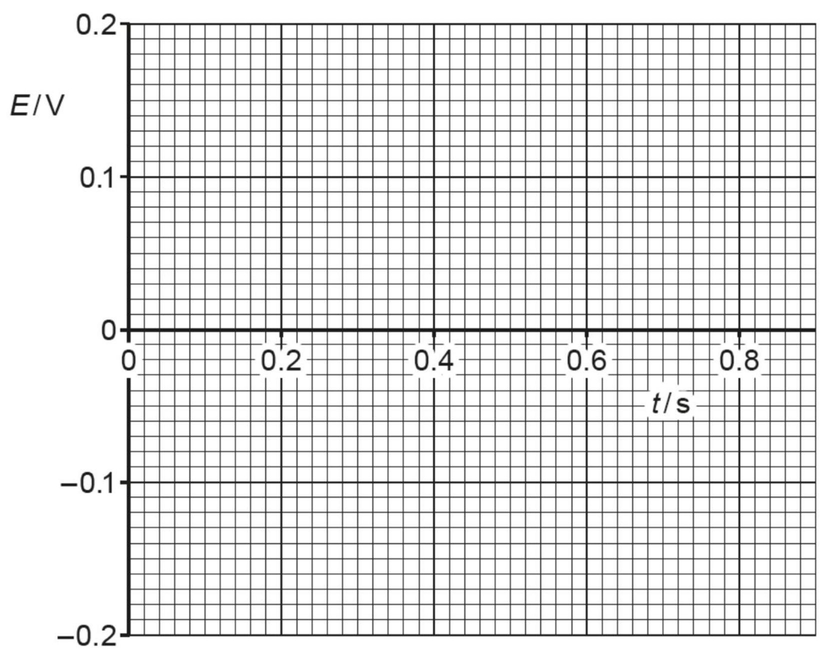

Fig. 10.5 [3]

(iii) The flux density B is now kept constant. The terminals of the smaller solenoid are connected together. The smaller solenoid is then removed from inside the larger solenoid.

Using laws of electromagnetic induction, explain why a force is needed to remove the smaller solenoid.

....

....

....

....

....

.... [3]

End of paper